SpeedCare: AI-Powered Medical Triage System

Project Status: Fully Functional Prototype | **Last Updated:** May 2026 | **Educational / Research Use**

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1. Project Overview

SpeedCare is an intelligent medical triage web application designed to streamline hospital patient intake. The system addresses hospital crowding, long wait times, and data entry errors through a dual-platform approach: a patient-facing mobile/kiosk interface and a comprehensive hospital staff portal.

The system operates on a **decision-support model** where patients provide personal details and a reason for their visit, while clinical decisions remain exclusively with hospital staff. The AI acts as a suggestion tool for personnel, never as a replacement for human judgment.

Core Objectives

- **Reduce Hospital Crowding:** Patients register digitally and wait seated, eliminating queues at service counters.
- **Language Accessibility:** Bilingual interface supporting both **English and Bisaya**.
- **Cognitive Ease:** Optimized for users under stress, including the elderly or injured.
- **Digital Records:** Eliminates paper-based registration and reduces manual data entry errors.
- **Staff Empowerment:** Personnel have real-time control over patient assignments and queue priority.
- **Responsible AI:** AI suggests triage levels, but humans make the final clinical decision.

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2. Design Philosophy: Cognitive Ease

SpeedCare is built on three pillars to ensure usability for all patients, regardless of technological familiarity:

- **Visual Clarity:** Uses a universal "traffic light" color-coding system (Red, Orange, Green, Blue) with large 44px touch targets and high-contrast text for readability.
- **Psychological Reassurance:** Provides constant feedback through progress indicators and real-time status updates.
- **Local Context:** Includes Bisaya medical terms (e.g., *"Sakit sa dughan"* for chest pain) and automatically prioritizes senior citizens (60+) to align with Filipino legal requirements.

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3. The 5-Step SpeedCare Flow

Step 1: Entry and Registration

Patients scan a QR code at the hospital gate using their phone or use a **dedicated hospital kiosk**.

- **Inputs:** Full name, age, and "Reason for Visit" (e.g., Consultation, Emergency, Laboratory).
- **Priority:** The system auto-detects priority status for Seniors (60+) and PWDs.
- **Output:** A queue reference number and a session QR code are generated.

Step 2: Staff-Led Assignment

The submission appears instantly on the Staff Portal dashboard.

- **AI Suggestion:** The AI analyzes the reason for visit and patient data to suggest a triage level (0–3).
- **Human Confirmation:** A staff member must manually confirm or override the suggestion; the patient never interacts with the AI directly.
- **Assignment:** Staff assigns a triage level, doctor, and room number.

Step 3: Real-Time Tracking & Hardware Checkpoints

Patients monitor their status via their smartphones or public display boards.

- **Mobile Tracking:** Shows "Submitted," "Under Review," "Assigned," or "Your Turn".
- **Wall-Mounted Checkpoints:** For patients without phones, **small scanning machines** are located in various corners of the hospital. Patients can scan their QR code to see if it is "almost their turn" or if they should remain in the waiting area.

Step 4: Digital Doctor Sync

The doctor reviews the patient's digital file on their workstation before the patient enters.

- **Digital Workflow:** After the checkup, the doctor pushes billing and laboratory requests to the next station digitally.
- **Efficiency:** This eliminates paper forms and handwritten notes between departments.

Step 5: QR Exit and Payment

The patient presents their Transaction QR code at the cashier.

- **Finalization:** The cashier scans the code to see an itemized bill.
- **Privacy:** Once payment is complete, session data is cleared to protect patient privacy.

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4. Priority Queue Logic

SpeedCare uses a three-rule sorting system to ensure fairness and clinical safety:

Rule	Category	Description

1	**Triage Level**	Level 0 (Red — Emergency) always moves to the front, regardless of arrival time.
2	**Priority Flag**	Seniors (60+) and PWDs are served before standard patients within the same triage level.
3	**Arrival Time**	Ties between patients with the same triage and priority status are broken by the earliest arrival timestamp.

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5. Hospital Info Chatbot

An optional, bilingual chatbot is available from the waiting screen to answer administrative questions.

- **Scope:** Provides info on department locations, doctor schedules, operating hours, fees, and required documents.
- **Safety Guardrails:** The chatbot **cannot** give medical advice, diagnose symptoms, or assess triage levels.
- **Emergency Protocol:** If a user mentions a medical concern, the bot directs them to wait for a doctor or seek immediate staff assistance.

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6. Technical Architecture & Roles

- **Frontend:** React 18.3.1 with TypeScript and Tailwind CSS v4.
- **Hardware:** Supports "Bring Your Own Device" (BYOD) via QR, standalone kiosks for registration, and wall-mounted scanning machines for queue checks.
- **Staff Roles:** Access is divided among Triage Nurses, Medical Coordinators, Doctors, and Admins to ensure secure data handling.
- **Audit System:** Every AI override is logged with a staff ID and timestamp for liability protection.